

DAWEI GUO

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Education

Nankai University, Tianjin, China

Sep 2021 - Expected Jun 2025

Bachelor of Internet of Things Engineering

Rank: 1/33 GPA: 3.83/4.0 (91.11/100)

- **Core Courses:** Computer Architecture, Computer Networks, Algorithms Design and Analysis, Database Systems, Operating System, Principles of Compilers, Embedded Systems, Machine Learning, etc.

Research Expeirence

Resource Scheduling Mechanisms in Multi-Tenant Database Caching Systems (*Team Member*) **2024 - Present**

- Designed and developed a testbench by rewriting the Yahoo! Cloud Serving Benchmark (YCSB) in C++ and integrating a unified cache layer across multiple backends, including MySQL, MongoDB, LevelDB, SQLite, and TMDb.
- Built YCSB workloads with diverse query distributions to accurately assess performance, and tested various cache structures, such as query cache and key-value cache, to evaluate their effectiveness in different scenarios.
- Analyzed performance metrics to inform further optimization and dynamic scheduling of resources, such as cache, CPU cores, and I/O resources.

ChebyKAN: Kolmogorov-Arnold Networks using Chebyshev polynomials. (*Personal Project*) **2024 - Present**

- Developed Kolmogorov-Arnold Networks (KAN) using Chebyshev polynomials for function approximation.
- Conducted experiments comparing ChebyKAN's performance with MLP in function approximation and MNIST classification.
- Worked with contributors on Github to enhance performance and validate the model's theoretical foundations.

syc - SysY Language Compiler (*Team Member*) **2023 - 2024**

- Developed a high-performance compiler for SysY (a subset of the C language), utilizing Yacc and Lex for parsing and LLVM IR for intermediate representation.
- Implemented RISC-V backends, integrating various compilation optimizations to enhance the compiler's performance, including automatic parallelization, static branch prediction, block layout optimization, etc.
- Currently rewriting from C++ to Rust to adopt a multi-level IR similar to MLIR, aimed at expanding support for additional dialects and improving scalability.

yaTuner: A Compiler Auto-Tuning Tool (*Team Leader*) **2022**

- Developed an automated compiler tuning tool crafted in Python.
- Utilized statistical hypothesis testing and LinUCB, along with Bayesian optimization algorithms, to optimize compiler settings efficiently.
- Achieved a balance between multiple optimization objectives, notably execution time and code size, enhancing overall compiler performance.

Internships Expeirence

Terapines Technology (Wuhan) Co.,Ltd (*MLIR Team Member*) **May. 2024 - Present**

- Conducted research on the Mojo language compiler, contributing to language development and upstream improvements.
- Implemented Llama3 inference in Mojo, benchmarking Mojo's performance against other languages and frameworks.
- Contributed to the integration and optimization of Triton-CPU with MLIR frameworks, enhancing performance and compatibility for RISC-V backends.

Honors and Awards

Nankai University GongNeng Scholarship (top 5% in Nankai University) **2022**

China National Scholarship (top 1% in China) **2023**

Competitions

National Collegiate Operating System Design Competition, Honorable Mention **May 2022 - Aug 2022**

Nankai University NFS Intelligent Car Championship Junior Division, First Place **May 2022**

Tianjin Collegiate Student Cybersecurity (Catch The Flag) Competition, Third Prize **Aug 2022**

National Collegiate Information Storage Technology Competition, Third Prize **October 2022 - April 2023**

National Collegiate Compiler System Design Competition, First Prize **May 2024 - August 2024**

Technical Skills

- Familiar with C/C++ programming and Qt GUI development.
- Proficient in Python programming, machine learning, and PyTorch deep learning.
- Familiar with modern NN architectures, including CNNs, LSTMs, and Transformer models, as well as LLMs.
- Knowledgeable about LLVM and MLIR architectures, with familiarity in LLVM IR and MLIR.
- Proficient in Rust programming and Mojo programming language.
- **Languages:** Chinese (Native), English (Fluent, TOEFL 105 (R30 L29 S23 W23)).



Undergraduate Academic Transcript

Name: Guo Dawei			Student ID: 2112052		Admission Date: 2021.09		Length of Schooling: 4 Years				
Department: College of Cyberspace Security					Major: Internet of Things Engineering						
Course Title			Score	Credit	Grade point	Course Title			Score	Credit	Grade point
Fall 2021					Skating and roller skatingRoller skating ball			86	1.0	3.7	
Higher Mathematics Class AI			75	5.0	2.7	Probability and Mathematical Statistic			88	4.0	3.7
Education of Ideology and Morality and Introduction to the Law			86	2.5	3.7	Discrete mathematics			94	4.0	4.0
College English I A			96	2.0	4.0	Digital Logic			97	3.5	4.0
Linear Algebra			83	4.0	3.3	Essentials of the Modern History of China			88	2.5	3.7
Table Tennis Beginner's Class			84	1.0	3.3	Spring 2023					
High Level Language Program Design2-1			90	3.5	4.0	Advanced English			94	2.0	4.0
Spring 2022					Software Security			96	2.5	4.0	
Higher Mathematics Class AII			97	5.0	4.0	Database System			96	3.5	4.0
Basic Principles of Marxism			89	3.5	3.7	Introduction to Mao Zedong's Thoughts and the Theory of Socialism with Chinese Characteristics			86	3.5	3.7
Basic Physics Experiment			94	2.0	4.0	Fundamentals of Computer Organization			85	3.5	3.7
College Physics (1)			91	4.0	4.0	The Principle of Embedded System			92	3.5	4.0
College Life Navigation			P	2.0	/	Introduction to Artificial Intelligence			90	2.5	4.0
Guidance of Major cognition			P	1.0	/	Nature Observation and Ecological Photography			P	1.0	/
College English II A			93	2.0	4.0	Sensing Technology and Application			87	2.5	3.7
Fundamentals of Electric Circuits			89	3.5	3.7	Visual Technology Basis			96	2.5	4.0
Swimming Beginner's Class			97	1.0	4.0	Algorithm Design and Analysis			85	3.5	3.7
High Level Language Program Design2-2			94	2.5	4.0	History of Western Music			W	2.0	/
Military Theory			91.5	2.0	4.0	Fall 2023					
Basic principle and simulation flight of UAV			P	2.0	/	Beautiful China Construction Documentary			97	2.0	4.0
Fall 2022					Network Technology and Application			91	2.5	4.0	
Python programming			98	2.5	4.0	Operating System			98	3.5	4.0
Preliminary Electronic Design			95	1.5	4.0	West African Music——Djembe & Dunun			P	1.0	/
The Italian Language			96	2.0	4.0	Computer Architecture			94	3.5	4.0
College Chinese			96	2.0	4.0	Project Training and Practice			96	2.0	4.0
Military skill training			98	2.0	4.0	Creation Basis and Appreciation of Film and Television Animation			89	1.0	3.7
Introduction to Internet of Things			93	2.0	4.0	Introduction to History			P	2.0	/
Data Structure			90	3.5	4.0	Introduction to XI Jinping thought on Socialism with Chinese Characteristics for the New Era			78	2.0	3.0
Assembly Language and Reverse Engineering			93	2.5	4.0	Tennis Beginner's Class			84	1.0	3.3
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Official Seal:

Issue Date: 2024-09-01



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Course Title	Score	Credit	Grade point	Course Title	Score	Credit	Grade point
Principles of Compilers	96	3.5	4.0				
Security of Internet of Things	85	2.5	3.7				
Computer Networks	96	3.5	4.0				
Spring 2024							
Global outlook and personal development	96.6	1.0	4.0				
Communication Technology of IoT	93	2.5	4.0				
Innovative Research and Training	P	1.0	/				
Intelligent Computing System	89	2.5	3.7				
---End of Transcript---							

南开大学本科考核成绩及计分方式说明

1. 南开大学本科教学课程考核方式分为考试和考查两种。
2. 考试成绩原则上均采取百分制记分,最高分 100,及格分 60。考查成绩以通过或未通过记录,并且不参与学分绩计算。
3. 平均学分绩是衡量学生学习质量的标准,平均学分绩计算方法如下:

$$\text{平均学分绩} = \frac{\text{所学课程学分绩(分数} \times \text{学分数)之和}}{\text{所学课程学分之和}}$$

4. 符号注释:

符号	注 释
P	通过
F	未通过
W	退课
I	未完成
△	校外交流获得成绩
*	全英文课程
M	慕课

注明:“通过”、“未通过”、“退课”、“未完成”及“校外交流获得成绩”均不参与学分绩计算。

Undergraduate Transcript Explanations of

Nankai University

1. There are two grading standards for undergraduate courses: examination and pass-fail pattern.
2. Examination courses take 100-point grading scale, with 100 as the maximum, and 60 as the lowest passing score. Pass-fail courses use P or F for grading, and they are not taken into account in the cumulative weighted average.
3. The cumulative weighted average is the measure of a student's academic performance.

$$\text{Cumulative weighted average} = \frac{\sum(\text{score} \times \text{credit})}{\sum \text{credit}}$$

score = score received in a course

credit = credit of a course

Σ = the sum of (for all taken courses)

4. Code explanation:

Code	Interpretation
P	Pass
F	Fail
W	Withdrawn
I	Incomplete
△	Credit Transfer
*	English Taught Course
M	MOOC

Courses marked with △ indicate credits transferred from other universities. The courses marked with P, F, W, I and △ are not taken into account in the cumulative weighted average.